Introduction to Real Analysis – Homework 6

Signed integrals and the Riemann integral

All functions are real-valued. Integrable means Lebesgue measurable with $\int |f|\,dm < \infty$; m denotes Lebesgue measure.

1. **Equality in the triangle inequality.** Let f be integrable on $\mathbb{R}^d$. Prove that

$$\left|\int_{\mathbb{R}^d} f\,dm\right| = \int_{\mathbb{R}^d} |f|\,dm$$

   if and only if $f \geq 0$ a.e. or $f \leq 0$ a.e.

2. **Testing on measurable sets.** Let f, g be integrable on $\mathbb{R}^d$.

   (a) Prove that $f \leq g$ a.e. if and only if

$$\int_E f\,dm \leq \int_E g\,dm \qquad \text{for every measurable } E \subset \mathbb{R}^d.$$

   (b) Deduce that $f = g$ a.e. if and only if their integrals over every measurable set agree. Give an example showing that equality of the integrals over $\mathbb{R}^d$ alone is insufficient.

3. **Integral and level-set estimates.** Let f_n, f be integrable on $\mathbb{R}^d$ and suppose

$$\int_{\mathbb{R}^d} |f_n - f|\,dm \longrightarrow 0.$$

   (a) Prove that

$$\int_{\mathbb{R}^d} f_n\,dm \longrightarrow \int_{\mathbb{R}^d} f\,dm, \qquad m\{|f_n - f| \geq \varepsilon\} \longrightarrow 0 \quad \text{for every } \varepsilon > 0.$$

   (b) Under this hypothesis, can uniform convergence fail even when $f_n(x) \to f(x)$ at every point? Give such an example on $\mathbb{R}$ with all functions zero outside $[0, 1]$.

4. **Changing a function on a null set.** Define

$$h(x) = \begin{cases} x + \mathbf{1}_{\mathbb{Q}}(x), & 0 \leq x \leq 1, \\ 0, & x \notin [0, 1]. \end{cases}$$

   (a) Show that h is Lebesgue integrable and compute $\int_{\mathbb{R}} h\,dm$.

   (b) Prove directly from Darboux sums that $h|_{[0,1]}$ is not Riemann integrable.